

Trigonometry

Igcse Mathematics

This handout covers Topic 6, Trigonometry. Parts marked **(Extended)** are only tested on the Extended papers; everything else is for both levels. Give angle answers to one decimal place.

Pythagoras' theorem

Pythagoras' theorem 勾股定理 links the three sides of a **right-angled triangle** 直角三角形 (a **triangle** 三角形 with one 90° angle). If the longest side (the **hypotenuse** 斜边, opposite the right angle) is c , then

$$a^2 + b^2 = c^2.$$

Use it to find a missing side.

Worked example. A right-angled triangle has a hypotenuse of 13 cm and one short side of 5 cm. Find the other short side.

$$b^2 = 13^2 - 5^2 = 169 - 25 = 144, \quad b = \sqrt{144} = 12 \text{ cm}.$$

Trigonometry in right-angled triangles

Label the sides from the angle you are using: the **opposite** 对边 (across from the angle), the **adjacent** 邻边 (next to the angle), and the hypotenuse. The three ratios are the **sine** 正弦, **cosine** 余弦 and **tangent ratio** 正切 ($\sin$, $\cos$, $\tan$):

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan \theta = \frac{\text{opp}}{\text{adj}}.$$

Remember them as **SOH-CAH-TOA**. To find an angle, use the inverse ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$).

Worked example (find a side). In a right-angled triangle the hypotenuse is 10 cm and the angle is 30° . Find the opposite side.

$$\text{opp} = 10 \times \sin 30^\circ = 10 \times 0.5 = 5 \text{ cm}.$$

Worked example (find an angle). The opposite side is 4 cm and the adjacent side is 3 cm.

$$\tan \theta = \frac{4}{3}, \quad \theta = \tan^{-1}\left(\frac{4}{3}\right) = 53.1^\circ.$$

Angles of elevation and depression (Extended)

The **angle of elevation** 仰角 is the angle up from the horizontal to an object above you. The **angle of depression** 俯角 is the angle down from the horizontal to an object below you. The shortest distance from a point to a line is the **perpendicular** 垂直 distance.

Worked example. From a point 50 m from the foot of a tower, the angle of elevation to the top is 40° . Find the height of the tower.

$$\text{height} = 50 \times \tan 40^\circ = 50 \times 0.839 = 42.0 \text{ m.}$$

Exact trigonometric values (Extended)

You must know these exact values without a calculator.

x	0°	30°	45°	60°	90°
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan x$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	—

Graphs and trigonometric equations (Extended)

For $0^\circ \leq x \leq 360^\circ$:

- $y = \sin x$ is a wave starting at 0, peaking at 90° , back to 0 at 180° , down to -1 at 270° .
- $y = \cos x$ is the same wave but starting at 1.
- $y = \tan x$ rises steeply and repeats every 180° .

A **trigonometric equation** 三角方程 often has more than one answer in this range. Use the graph (or the symmetry of the wave) to find them all.

Worked example. Solve $\sin x = \frac{\sqrt{3}}{2}$ for $0^\circ \leq x \leq 360^\circ$.

One answer is $x = 60^\circ$. The sine wave is also $\frac{\sqrt{3}}{2}$ at $180^\circ - 60^\circ = 120^\circ$. So $x = 60^\circ$ or 120° .

Worked example. Solve $2 \cos x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$.

$$\cos x = -\frac{1}{2} \Rightarrow x = 120^\circ \text{ or } 240^\circ.$$

The sine and cosine rules (Extended)

For **any** triangle (not just right-angled), with sides a, b, c opposite angles A, B, C :

$$\text{sine rule: } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C},$$

$$\text{cosine rule: } a^2 = b^2 + c^2 - 2bc \cos A.$$

Use the **sine rule** 正弦定理 when you have a side and its opposite angle. Use the **cosine rule** 余弦定理 when you have two sides and the angle between them, or all three sides. With the sine rule, watch for the **ambiguous case** 两解情况, where an angle could be acute or obtuse.

The area of any triangle is

$$\text{area} = \frac{1}{2}ab \sin C.$$

Worked example. A triangle has $b = 7$ cm, $c = 8$ cm and the angle between them $A = 40^\circ$. Find side a .

$$a^2 = 7^2 + 8^2 - 2(7)(8) \cos 40^\circ = 113 - 112 \times 0.766 = 27.2,$$

$$a = \sqrt{27.2} = 5.2 \text{ cm.}$$

Pythagoras and trigonometry in 3D (Extended)

In three dimensions, find a right-angled triangle inside the solid, then use Pythagoras or trigonometry on it. A common task is the angle between a line and a flat surface (a **plane** 平面).

Worked example. A box has a base 6 cm by 8 cm and height 5 cm. Find the angle between a space diagonal and the base.

First the base diagonal: $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$ cm. This diagonal and the height form a right-angled triangle, so the angle θ with the base is

$$\tan \theta = \frac{5}{10} = 0.5, \quad \theta = \tan^{-1}(0.5) = 26.6^\circ.$$